Lecture 9: Risk Aversion vs Intertemporal Substitution

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Intro

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Building Blocks

What is Risk Aversion?

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- You don't need the concept of *time* to think about risk aversion

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- You don't need the concept of *time* to think about risk aversion
- X: a set of outcomes; $\Delta(X)$: the set of lotteries over X (the simplex)
- Let $\succeq$ be a preference relation over $\Delta(X)$
- You are **risk-averse** if, for any $p \in \Delta(X)$, you have $\delta_{\mathbb{E}[p]} \succeq p$

What Von-Neumann and Morgenstern Taught Us

- Typical assumptions: $\succeq$ is complete, transitive, and continuous
- Independence axiom: if $p \succeq q$, then for any r and $\alpha \in (0,1)$, we have $\alpha p + (1-\alpha)r \succeq \alpha q + (1-\alpha)r$

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- Independence axiom: if $p \succeq q$, then for any r and $\alpha \in (0,1)$, we have $\alpha p + (1-\alpha) r \succeq \alpha q + (1-\alpha) r$
- Under independence, $\succeq$ can be represented by a utility function $U : \Delta(X) \to \mathbb{R}$ such that

$$U(p) = \sum_{x \in X} \pi(x) u(x)$$

  where $\pi(x)$ is the probability of outcome x and $u(x)$ is the utility of outcome x
- The preferences we will cover in this lecture will violate exactly this axiom, stay tuned!

More or Less Risk Averse?

- The **Certainty Equivalent** of $p \in \Delta(X)$ is the amount of money c such that $\delta_c \sim p$

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- John is (weakly) **more risk-averse** than Mary if $CE_J(p) \leq CE_M(p)$ for all $p \in \Delta(X)$
- Pratt showed that this is all about how concave u is!
- Concavity is measured by second derivatives... but these are not scale-free!
- We typically measure risk aversion by $-u''(.)/u'(.)$ or $-x \cdot u''(x)/u'(x)$
- Many prominent utility functions: CARA, CRRA, HARA, ...

The Special Case of CRRA

- In Macro, we often use the CRRA utility function:

$$u(C) = \frac{C^{1-\gamma} - 1}{1-\gamma}$$

- The parameter γ is the coefficient of relative risk aversion (RRA)

$$\frac{-c \cdot u''(c)}{u'(c)} = \gamma$$

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- If $x \succeq y$, then $\alpha x \succeq \alpha y$ for any $\alpha > 0$ (Indifference curves are radially symmetric)
- Optimal portfolio weights are independent of wealth (with complete markets)
- Equilibrium prices (and returns) depend on *growth rates*, not levels

Questions?

Intertemporal Substitution

- You don't need the concept of *uncertainty* to think about intertemporal substitution
- Think about two periods and a consumer choosing optimal consumption:

$$\max_{C_1, C_2} u(C_1) + \beta u(C_2) \quad \text{s.t.} \quad C_1 + \frac{C_2}{1+r} = W$$

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- **Elasticity of Intertemporal Substitution (EIS)**: $\frac{d\log(c2/c1)}{d\log(1+r)} > 0$
- Exercise: show that for CRRA preferences, the EIS is $1/\gamma$

Risk Aversion vs Intertemporal Substitution

- With CRRA preferences, high $\gamma \implies$ high risk aversion
- But high $\gamma \implies$ low EIS $\implies$ low willingness to substitute consumption
- Only two possibilities:
  - High γ: very risk-averse, hates shifting consumption around
  - Low γ: not very risk-averse, and very willing to shift consumption around

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  - Low γ: not very risk-averse, and very willing to shift consumption around
- If you bump up γ to make the consumer afraid of equity...
- She will really upset about shifting consumption around!
- You have to make the relative price of consuming tomorrow low $\implies$ high risk-free rate!
- The prize for waiting must be large... risk-free rate puzzle from Weil (1989)!

Questions?

A New Class of Preferences Comes to Town

Do We Care About *When* Uncertainty is Resolved?

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- The timing of uncertainty is irrelevant
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Do We Care About *When* Uncertainty is Resolved?

- With expected utility, only probabilities and outcomes matter
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- But what if you care about *when* uncertainty is resolved?
- An irreversible medical condition... do you want to find out now or later?
- At the slot machine in Vegas... do you want to wait and gamble, or to resolve it immediately?
- It's already decided whether you get tenured or not... do you want to find out now or later?

Kreps-Porteus Preferences

- Kreps and Porteus (1978): axiomatic characterization of preferences over lotteries *and* timing
- In the EU framework, "time" only matters because it induces other *states*...
- Example: flip a coin, consume 5 at $t = 0$, 10 at $t = 1$ if heads, 0 at $t = 1$ if tails
- Do you want to find out the outcome of the coin flip at $t = 0$ or at $t = 1$?

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- Under EU, you should be indifferent between options
- KP preferences allow you to prefer one over the other
- EU: preferences over probability distributions
- KP: preferences over "filtrations" – information structures
- KP recover EU when the agent does not care about timing

Quick Look Into Their Setup (Skipping Super Technical Terms)

- Time is discrete $t \in \{0, 1, 2, \dots, T\}$
- For every t, there is a payoff space Z_t
- D_T is the set of probability measures over Z_T (lotteries at time T)

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- $D_{T-1} \equiv Z_{T-1} \times X_T$

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- In general: $D_t \equiv Z_t \times X_{t+1}, \forall t < T$
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- In general: $D_t \equiv Z_t \times X_{t+1}, \forall t < T$
- Your preference relation $\succeq$ ranks elements of D_t for all t
- Usual stuff: completeness, transitivity, continuity, monotonicity, ...
- If you want some stuff, check the paper!

Example

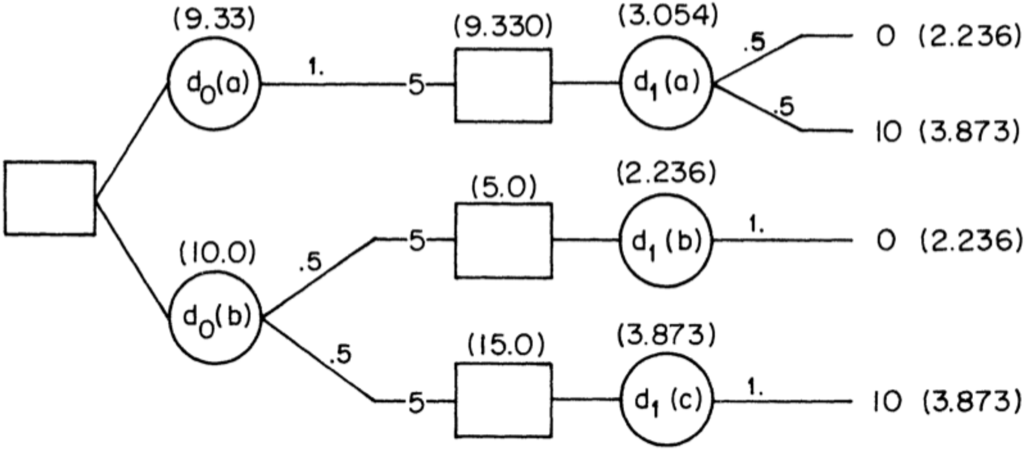

FIGURE 2.

The Representation Theorem

- In a finite decision tree, suppose U_t is your continuation value
- U_{t+1}: the *random* (from the point of view of t) continuation value tomorrow
- Let $\mu_t(U_{t+1})$ be the certainty equivalent of U_{t+1}
- Interpretation: money you get right now, for sure, to stop playing

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- Interpretation: money you get right now, for sure, to stop playing
- Axioms from Kreps and Porteus (1978) $\implies$ there exists an "aggregator" function f such that

$$U_t = f(Z_t, \mu_t(U_{t+1}))$$

Intuition

- $\mu_t(.)$ encodes how risk-averse you are: the more risk-averse, the lower $\mu_t(U_{t+1})$ is
- The aggregator measures how much you trade-off payoff today vs keep playing
- But the continuation value is already summarized by $\mu_t(.)$

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- The aggregator measures how much you trade-off payoff today vs keep playing
- But the continuation value is already summarized by $\mu_t(.)$
- If f is linear in the second argument and $\mu_t(U_{t+1}) = \mathbb{E}_t[U_{t+1}]$ we recover expected utility!

Epstein and Zin (1989)

- L. Epstein and Zin (1989) made the whole setup operational
- They extended Kreps and Porteus (1978) to the infinite horizon case
- Super non-trivial since there is no "end date"... continuation becomes a fixed point problem

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- They extended Kreps and Porteus (1978) to the infinite horizon case
- Super non-trivial since there is no "end date"... continuation becomes a fixed point problem
- They specialize functional forms:

$$f(x,y) \equiv \left((1-\beta)x^{1-\rho} + \beta y^{1-\rho}\right)^{\frac{1}{1-\rho}}$$
$$\mu_t(U_{t+1}) \equiv \left(\mathbb{E}_t[U_{t+1}^{1-\alpha}]\right)^{\frac{1}{1-\alpha}}$$

- α controls risk aversion, ρ controls intertemporal substitution

Heuristic Derivation of the SDF

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- L. G. Epstein and Zin (1991) took it to the data and showed how to use the implied SDF
- We will follow a heuristic derivation from François Gourio's lecture notes (see Github)
- We will assume, just to make math clear, that we have a countable state space
- Expectations will become sums weighted by conditional probabilities $\pi_t(s_{t+1})$

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$$\mu_t(U_{t+1}) = \left(\sum_{s_{t+1}} \pi_t(s_{t+1}) \, U_{t+1}(s_{t+1})^{1-\alpha} \right)^{\frac{1}{1-\alpha}}$$

The Recursion

- Substituting μ_t into f, the Epstein-Zin recursion is:

$$U_t = \left[(1-\beta) C_t^{1-\rho} + \beta \left(\sum_{s_{t+1}} \pi_t(s_{t+1}) \, U_{t+1}(s_{t+1})^{1-\alpha} \right)^{\frac{1-\rho}{1-\alpha}} \right]^{\frac{1}{1-\rho}}$$

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- Let $q_t(s_{t+1})$ be the time-t price of an Arrow-Debreu security
- If you buy one unit, your utility changes by $\partial U_t / \partial C_{t+1}(s_{t+1})$
- You pay $q_t(s_{t+1})$ and each unit of consumption hurts you by $\partial U_t / \partial C_t(s_t)$

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- You pay $q_t(s_{t+1})$ and each unit of consumption hurts you by $\partial U_t / \partial C_t(s_t)$
- At the margin $\dfrac{\partial U_t}{\partial C_{t+1}(s_{t+1})} = q_t(s_{t+1}) \cdot \dfrac{\partial U_t}{\partial C_t(s_t)}, \forall s_{t+1}, s_t, t$
- For any asset i with return $R_{i,t+1}(s_{t+1})$ in state s_{t+1}, we have

$$\mathbb{E}_t\left[M_{t+1}\, R_{i,t+1}\right] = \sum_{s_{t+1}} \pi_t(s_{t+1})\, M_{t+1}(s_{t+1})\, R_{i,t+1}(s_{t+1}) = \sum_{s_{t+1}} q_t(s_{t+1}) \cdot R_{i,t+1}(s_{t+1}) = 1$$

Marginal Utilities

- Raise both sides of the recursion to the power $1-\rho$:

$$U_t^{1-\rho} = (1-\beta)\, C_t^{1-\rho} + \beta\, \mu_t^{1-\rho}$$

- Differentiate w.r.t. C_t:

$$\frac{\partial U_t}{\partial C_t} = (1-\beta)\left(\frac{C_t}{U_t}\right)^{-\rho}$$

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- For $C_{t+1}(s_{t+1})$, chain rule: $\dfrac{\partial U_t}{\partial C_{t+1}(s_{t+1})} = \dfrac{\partial U_t}{\partial \mu_t} \cdot \dfrac{\partial \mu_t}{\partial U_{t+1}(s_{t+1})} \cdot \dfrac{\partial U_{t+1}(s_{t+1})}{\partial C_{t+1}(s_{t+1})}$

$$\frac{\partial U_t}{\partial \mu_t} = \beta \left(\frac{\mu_t}{U_t}\right)^{-\rho}, \qquad \frac{\partial \mu_t}{\partial U_{t+1}(s_{t+1})} = \pi_t(s_{t+1}) \left(\frac{U_{t+1}(s_{t+1})}{\mu_t}\right)^{-\alpha},$$

$$\frac{\partial U_{t+1}(s_{t+1})}{\partial C_{t+1}(s_{t+1})} = (1-\beta) \left(\frac{C_{t+1}(s_{t+1})}{U_{t+1}(s_{t+1})}\right)^{-\rho}$$

The SDF

The ratio of marginal utilities yields:

$$q_t(s_{t+1}) = \frac{\partial U_t / \partial C_{t+1}(s_{t+1})}{\partial U_t / \partial C_t(s_t)} = \beta \pi_t(s_{t+1}) \left(\frac{C_{t+1}(s_{t+1})}{C_t} \right)^{-\rho} \left(\frac{U_{t+1}(s_{t+1})}{\mu_t} \right)^{\rho-\alpha}$$

- We are left with:

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- When $\alpha = \rho$: the last term vanishes and we recover standard CRRA

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\rho}$$

- Problem: $U_{t+1}(s_{t+1})/\mu_t$ is not directly observable
- We will deal with this later using the return on the “wealth portfolio”

Questions?

The Wealth Portfolio Trick

Meeting the Data

Can Kreps-Porteus Preferences Solve the Equity Premium Puzzle?

- Weil (1989): no, not really. You need way more than only that

Net risk premium and net risk-free rate[a] ($\beta = 0.98$)

EIS $(1/\rho)$	CRRA (γ) 0	0.5	1	5	10	20	45
∞	0.00 **2.04**	0 05 **2.02**	0.09 **1.99**	0.47 **1.80**	0.93 **1.58**	1.76 **1.21**	3 00 **0.91**
2	0.01 **2.95**	0.06 **2.91**	0.11 **2.87**	0.51 **2.55**	1.00 **2.17**	1.88 **1.53**	3 13 **0.75**
1	0.01 **3.86**	0 07 **3.81**	0 12 **3.75**	0.55 **3.31**	1 07 **2.77**	2 00 **1.85**	3 27 **0.59**
0.2	0.10 **11.49**	0.18 **11.29**	0.26 **11.10**	0 89 **9.56**	1.65 **7.72**	2.93 **4.45**	4 37 **−0.68**
0 1	0 25 **21.87**	0 36 **21.47**	0 47 **21.08**	1.33 **17.95**	2 37 **14.26**	4 10 **7.83**	5.79 **−2.24**
0.05	0.59 **45.89**	0.75 **44.95**	0.89 **43.98**	2.18 **36.92**	3 73 **28.72**	6 37 **15.01**	8 82 **−5.22**
1/45	1 19 **131.52**	1 42 **128.87**	1.66 **125.06**	3.70 **101.02**	6 40 **75.11**	11.50 **36.06**	17.53 **−12.00**

[a] In percent The bold number is the net risk-free rate

The Way Forward

- There are two more fundamental reasons for the puzzle:
    - There is not enough aggregate risk in consumption growth
    - The average consumption growth is too high to be consistent with low interest rates

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    - There is not enough aggregate risk in consumption growth
    - The average consumption growth is too high to be consistent with low interest rates
- Increasing risk in consumption growth: disasters, long-run risks, ...
- Or introduce very imperfect insurance markets (e.g. incomplete markets, borrowing constraints, ...)
- Heterogeneity across agents is likely important!

Readings and References i

- Campbell, Chap. 6

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